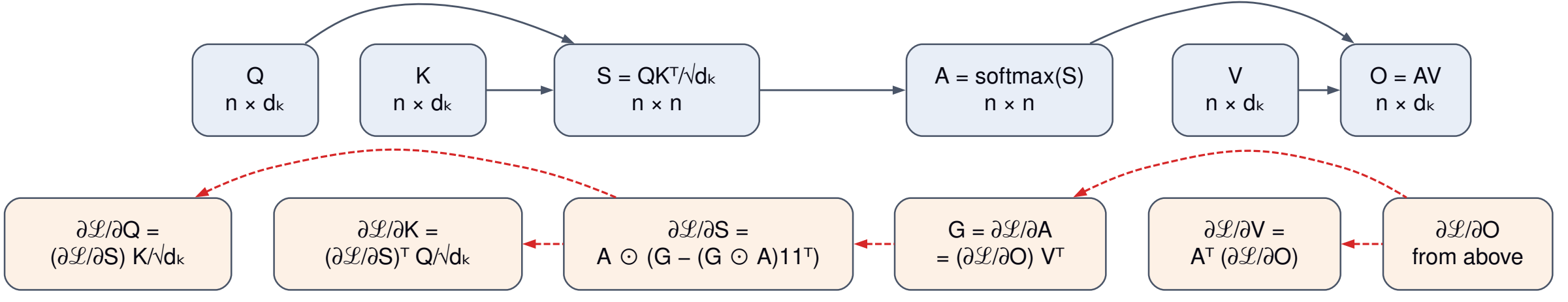
Q
n × dk
K
n × dk
S = QKT/√dk
n × n
A = softmax(S)
n × n
V
n × dk
O = AV
n × dk
∂ℒ/∂Q =
(∂ℒ/∂S) K/√dk
∂ℒ/∂K =
(∂ℒ/∂S)T Q/√dk
∂ℒ/∂S =
A ⊙ (G – (G ⊙ A)11T)
G = ∂ℒ/∂A
= (∂ℒ/∂O) VT
∂ℒ/∂V =
AT (∂ℒ/∂O)
∂ℒ/∂O
from above